Project 2: Automation of Concrete Mix Design
Written Summary

Jessica Tackett, Tessa Soukup, Addison Jehorek, Jamie Krab

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Introduction/Background:

NDOT has chosen our firm to convert their Excel-based concrete mix design model into a Python-based system. The main goal is to replicate the logic in the Excel "Mix Design" worksheet into python. This will allow users to input design parameters and material properties in a clean and structured manner. This will then create an output that generates a weight chart for one cubic yard of concrete. Migrating the Excel model to Python will allow for future integration into a web-based tool.

To further understand this concrete mix design, our team conducted some research into the materials or inputs we work with. One of the main inputs in our code, or weight chart, is fly ash and silica fume. These materials actually improve the mechanical and durability properties of concrete. Fly is a waste product that comes from burning coal. By replacing a percentage of the cement's weight with fly ash and silica fumes leads to strengthening in the cement concrete. Finding out a use for fly ash has been beneficial as it is sustainable for the coal business. If left unused, the fly ash is dumped on vast tracts of land, harming the environment and those living nearby. Silica fume is a mineral additive used to create sustainable concrete (Anish et al., n.d.). This material is derived from the production of silicon or ferrosilicon (Cemex US, n.d.). The benefits of using silica fume include lower CO2 emissions, more affordable concrete, increased durability, and better mechanical qualities. As for the other SCM input, SCM stands for supplementary cementitious materials (SCM). Concrete is usually a mixture of Portland cement, sand, coarse aggregate, and water. Portland cement is the binding agent. SCM can be a partial replacement of Portland cement to improve durability, strength, and sustainability goals. Fly ash and silica fume fall under SCMs in concrete mixture, but there are many other materials that also fall into this category. This includes, but is not limited to, calcined clay, ground glass, and natural pozzolans. All in all, these materials help reduce the environmental footprint of concrete construction (Cemex US, n.d.). As for the water to cement ratio, this is necessary in order to ensure optimal strength and workability. The recommended range is between 0.40-0.60(Water to Cement Ratio, n.d.).

The aggregates that are inputted for the weight chart all also have their own unique purposes. Fine aggregates, for example, influence the workability, finish, and durability of concrete. Some examples of fine aggregates include sand, manufactured sand, and crushed stone sand. Coarse aggregates are large particles that form the skeleton of concrete. Examples of this are crushed stone, gravel, boulders, and even recycled concrete. This aggregate provides resistance to wear and weathering, better load-bearing capacity, and strength (Kushwaha, 2025). Other aggregates can include natural aggregates, artificial aggregates, and recycled aggregates.

Methods:

The automation of Excel into Python-based software can be split into four tasks.

The first task is to translate the Excel logic into Python. To do so, our group will replicate the Excel calculations in Python using functions and variables. Each Excel calculation will require its own function and the variables within the function will be defined by the user. The Excel sheet was reviewed by group members to find each necessary variable and the formulas involved. Below is a figure of two functions used in Excel replicated in our python code.

```
#Functions
#For cement volume:
def vol_cement(w_cement, sg_cement):
    vol_R = w_cement / (sg_cement * unit_weight_water)
    return vol_R

#For fly ash volume
def fly_ash_vol(w_flyash, sg_flyash):
    vol = w_flyash /(sg_flyash * unit_weight_water)
    return vol
```

The second task is to implement sequential user inputs. This will be done by programming the code to accept user inputs using the input() function in Python. The input can be made to be an integer, float, or string. Material components and parameters will be float variables while the general information will be string variables. When all the inputs have been entered correctly, the user will be notified that they can run the next code. Below is the two types of user inputs described.

```
#user inputs
#Project metadata
project_number = int(input("Enter project number: "))
class_of_concrete = input("Enter class of concrete: ")

#Cementitious material inputs (lb per cubic yard)
cement_A = float(input("Enter cement weight A (lb per cubic yard): "))
fly_ash_B = float(input("Enter fly ash weight B (lb per cubic yard): "))
```

Once these inputs have been completed, the user can run the calculations. The third task is to generate a final mix design output to organize the results of these calculations. This will be done by creating a table in Python using print() functions. This table will display a weight chart for one cubic yard of concrete. This will be labeled and formatted so that the user can easily read results. Below is part of the code used to print a readable results chart.

```
print("\n---------------------------------------------")
print(" NDOT Concrete Mix Design – Weight Summary")
print("         (1 Cubic Yard of Concrete)")
print("---------------------------------------------")
print(f"Project Number:       {project_no}")
print(f"Class of Concrete:    {concrete_class}")
print("---------------------------------------------")
print(f"Cement (A):            {cement_A:8.1f} lb")
print(f"Fly Ash (B):           {fly_ash_B:8.1f} lb")
```

To check that these tasks are completed and that this Python code is successfully running like the NDOT Excel, the code will be evaluated using four concrete mix scenarios. The four concrete mix scenarios were researched and documented in the code. The scenarios followed the same python logic previously described.

Mix Scenarios Research and Documentation

Scenario #1 47B-3500

47B-3500 is a class of concrete that has a compressive strength of 3500 psi. The other concrete classes will follow this same structure, with the compressive strength coming at the end of the name (Nebraska Department of Transportation, n.d-a). This type of concrete is designed to meet the demands of construction projects such as roads, sidewalks, and infrastructure. Portland cement is used in this class of concrete (Nebraska Department of Transportation, n.d-b).

Scenario #2 Flowable Fill

Flowable fill concrete is a self-compacting and fast settling concrete used for backfilling voids, trenches, and structures. It eliminates the need for compaction equipment and reduces labor. The concrete flows into tight spaces and sets quickly (Flashfill Services, 2025). Flowable fill is a mixture of cement, fly ash, fine sand, water, and air (City of Lincoln, n.d.-a). The approximate compressive strength should range from 85 to 175 psi. No less than 95% of sand should pass through the No.4 sieve which is under the category of fine aggregate. No more than 5% should pass through the No.200 sieve which is the finest sieve.

Scenario #3 L-3500

L-3500 is a class of concrete that is a specific type of Portland Cement Concrete (PCC). Its general use is for pavements, sidewalks, and structures. This class of concrete is usually used in areas of high compressive strength being able to withstand significant loads (City of Lincoln, n.d.-b). The minimum PSI strength for this mixture is 3500 psi.

Scenerio #4 LB-2750

LB-2750 is a class of concrete that is used for new construction of residential streets (City of Lincoln, n.d.-c). This is generally a pavement base with the minimum PSI being 2750.

Results:

The code was tested using four different concrete classes to evaluate its reliability across a range of NDOT mix designs. The classes included 47B-3500, Flowable Fill, L-3500, and LB-2700 concrete. The results of these four scenarios are summarized below. Cementitious weights, specific gravities, and aggregate percentages were taken from the Standard Specifications for Highway Construction published by the Nebraska Department of Transportation (Nebraska Department of Transportation, 2020) and Standard Specifications for Highway Construction published by the City of Lincoln, (City of Lincoln, 2020).

Table 1. Class 47B-3500 Results

```
---------------------------------------------
NDOT Concrete Mix Design – Weight Summary
      (1 Cubic Yard of Concrete)
---------------------------------------------
Project Number:       2
Class of Concrete:    47B-3500
---------------------------------------------
Cement (A):            470.0 lb
Fly Ash (B):           120.0 lb
Silica Fume (C):         0.0 lb
Other SCM (D):           0.0 lb
---------------------------------------------
Fine Aggregate (Y):       1186 lb
Coarse Aggregate (Z):     1779 lb
Other Aggregate (AA):        0 lb
---------------------------------------------
Water (Q):              266 lb
---------------------------------------------
End of Mix Design Summary
```

Table 2. Flowable Fill Results

```
---------------------------------------------
NDOT Concrete Mix Design – Weight Summary
      (1 Cubic Yard of Concrete)
---------------------------------------------
Project Number:       2
Class of Concrete:    Flowable Fill
---------------------------------------------
Cement (A):             60.0 lb
Fly Ash (B):           200.0 lb
Silica Fume (C):         0.0 lb
Other SCM (D):           0.0 lb
---------------------------------------------
Fine Aggregate (Y):       3437 lb
Coarse Aggregate (Z):        0 lb
Other Aggregate (AA):        0 lb
---------------------------------------------
```

Water (Q):   117 lb

End of Mix Design Summary

Table 3. Class L-3500 Results

NDOT Concrete Mix Design – Weight Summary
(1 Cubic Yard of Concrete)

Project Number: 2
Class of Concrete: L-3500

Cement (A):   564.0 lb
Fly Ash (B):   0.0 lb
Silica Fume (C):   0.0 lb
Other SCM (D):   0.0 lb

Fine Aggregate (Y):   2083 lb
Coarse Aggregate (Z):   893 lb
Other Aggregate (AA):   0 lb

Water (Q):   282 lb

End of Mix Design Summary

Table 4. Class LB-2750 Results

NDOT Concrete Mix Design – Weight Summary
(1 Cubic Yard of Concrete)

Project Number: 2
Class of Concrete: LB-2700

Cement (A):   423.0 lb
Fly Ash (B):   0.0 lb
Silica Fume (C):   0.0 lb
Other SCM (D):   0.0 lb

Fine Aggregate (Y):   1901 lb
Coarse Aggregate (Z):   1267 lb
Other Aggregate (AA):   0 lb

Water (Q):   254 lb

End of Mix Design Summary

47B-3500 is used for pavement, curbs, gutters, sidewalks, and driveways. Flowable Fill is concrete used for filling trenches and structures. L-3500 is used for pavements, sidewalks, and structures. It is able to withstand significant loads. LB-2750 is used for residential streets. Flowable Fill uses the greatest weight of fine aggregate. This makes sense as it needs to fill voids and requires the flow to be able to fill random sized gaps. The LB-2750 and 47B-3500 have the most similar results, which also makes sense as they both have very similar uses.

Based off the results of the scenarios, the code correctly determined the concrete weights and parameters for each class. The user should be able to enter in their concrete of choice and receive accurate results.

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